MOHAMMAD MAHAD NADEEM

M.Sc. Electrical Engineering
Senior Electrical Engineer

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SUMMARY

Senior Electrical Engineer with an extensive background in power electronics, AC/DC system design, and end-to-end engineering of energy conversion systems. Experienced in designing SLDs, protection/coordination schemes, wiring diagrams, and control architecture for converters, EV chargers, and power distribution systems. Strong expertise in hardware design, converter topology, control algorithm development, and system-level integration. Proven track record in testing, troubleshooting, and commissioning complex electrical systems while ensuring compliance with IEC/NEN standards. Adept at collaborating with international, cross-functional teams to deliver robust, efficient, and scalable energy solutions.

WORK EXPERIENCE

- **EWT Systems B.V.**
  Amersfoort, Netherlands
  April 2026 – Present

Senior Electrical Engineer

- Design and deliver end-to-end electrical control panels and systems, including architecture, component sizing, schematics, BOMs, and full documentation packages compliant with IEC standards.
- Integrate LV/MV power systems, generators, MV transformers, grid connection, coordinating protection schemes, earthing, and EMC across multi-discipline teams.
- Supported power system studies (load flow, short-circuit, harmonics) and led validation activities including DFMEA, FAT/SAT, and root-cause analysis.
- Prepared technical documentation for certification, audits, and customer submissions, ensuring full traceability from requirements to design and test evidence.

- **Schneider Electric (DC Systems)**
  Aalsmeer, Netherlands
  July 2025 – March 2026

Senior System Engineer

- Lead end-to-end execution of electrical system projects from initial design through commissioning, managing scope, schedules, resources, and technical coordination with internal and external stakeholders.
- Design Single Line Diagram (SLDs), protection/control schematics, wiring diagrams, I/O lists, and panel layouts.
- Translate customer and project requirements into technically compliant, cost-effective electrical solutions aligned with IEC/NEN standards and Schneider design guidelines.
- Collaborate with cross-functional engineering teams (firmware, mechanical) to ensure complete system integration and interface compatibility.
- Perform system-level testing, troubleshooting, FAT/SAT support, and root-cause analysis to optimize electrical system performance and resolve technical issues deployed systems.
- Prepare and maintain engineering documentation, technical reports, and design approval packages for internal reviews and customer sign-off.

February 2025 – June 2025

System Engineer

- Responsible for designing and developing Single Line Diagrams (SLDs).
- Supported integration of power systems and DC products into operational environments.
- Assisted in project planning, system testing, and validation of hardware/ software components.

June 2024 – January 2025

Power Electronics Engineer

- Leading the design and management of High Voltage DC products (350V DC & 700V DC) from concept to deployment.
- Overseeing hardware and system design, ensuring compliance with industry standards and project requirements.
- Coordinating and executing product testing, validation, & performance analysis.

- **TalTech**
  Tallinn, Estonia
  February 2023 – May 2024

Project Researcher

Project:

Universal Battery AC/DC Charger for Electric Vehicles

- Simulation, modelling and designing the EV converter (from scratch).
- Topology selection/design.
- Control design of EV charger.

- **SkyElectric Pvt Ltd.**
  Islamabad, Pakistan
  October 2021 – June 2022

Project Design & System Integration Engineer

Project:

Designing hardware and control system of single-phase inverter in Grid-Tie, Stand-Alone and Hybrid modes for 5.5kW system for Japan:

- Designed the hardware of the inverter.
- Provide technical expertise in problem resolution, cost reduction and quality improvement.
- Designed the control using the Stateflow in MATLAB.
- Designed the anti-islanding algorithm for implementing the Japanese Electrical Manufacturer Standard – 1498.

- **Soongsil Univeristy**
  Seoul, South Korea
  September 2019 – June 2021

Research Assistant

Research/Project:

1.) A Robust Harmonic Elimination Method for Three Phase Grid-Connected Inverter Using Digital Lockin-Amplifier:

- Designed the LCL filter for 10kW inverter meeting the IEEE Std.519 & 1547.
- Designed PCB for the inverter voltage and current sensing circuits.
- Designed SOGI & Frequency Locked Loop based Phase Locked Loop (PLL) for synchronization to grid and harmonic elimination.
- TMS320F28335 DSP code written in Code Composer Studio for inverter control.

2.) A Hybrid Anti-Islanding Algorithm Using Digital Lock-In Amplifier:

- Designed algorithm for islanding detection according to IEEE Std. 1547.
- Designing of non-invasive sensor less anti-islanding algorithm for inverters
- Cost effective, simple & robust.

3.) A Multi-Channel Fast Impedance Spectroscopy Instrument for Quality Assurance of Super-Capacitors:

- A fast Electrochemical Impedance Spectroscopy (EIS) methodology proposed for quality assurance of the supercapacitors.

- A robust methodology instrument.

EDUCATION

- **Soongsil University**
  Seoul, South Korea
  Sept 2019 – Aug 2021

Master's in Electrical Engineering
- **Major:** Power Electronics

Thesis Title: "A Hybrid Anti-Islanding Algorithm Using Digital Lock-in Amplifier"
Keywords: Electronics, Research, Inverters, Converters, Renewable Energy, Power Quality, Super-Capacitors, Product Development

SKILLS

- **Software** - PSIM - Proteus - SEE Electrical - Eplan - Altium - Solidworks - MATLAB - LTspice
- **Programming** - C++

LANGUAGES

- **Dutch** Beginner (*Actively learning*)
- **English** Advanced
- **Urdu** Native

PROFESSIONAL CERTIFICATIONS & AWARDS

- **NEN 1010**
  March 2025
  Awarded by Royal Dutch Standardization Institute
  - Awarded training certification
- **NEN 3140 Skilled Person**
  March 2025
  Awarded by Royal Dutch Standardization Institute
  - Awarded training certification
- **NPR 9090 DC-installations**
  March 2025
  Awarded by Royal Dutch Standardization Institute
  - Awarded training certification
- **International Scholarship**
  Sept 2019 to Aug 2021
  Awarded Scholarship by Soongsil University
  - Awarded scholarship 100% by Soongsil University during Masters.
- **IGNITE ICT Research & Development Fund**
  Aug 2017 to May 2018
  Project: Stewart Platform for Dynamic Stabilization
  - Awarded National R&D fund by Government of Pakistan.